

3SAT $\longrightarrow$ zero-finding with constant Jacobian

CLAIM

For every 3CNF formula ϕ with n variables and m clauses, one can construct a degree-7 polynomial map $K_\phi : \mathbb{R}^{3n+3m} \rightarrow \mathbb{R}^{3n+3m}$ such that

$$K_\phi^{-1}(0) \neq \emptyset \iff \phi \in \text{SAT}, \quad \det JK_\phi \equiv (-3456)^{n+m}.$$

- 1. Start with a machine that has two special targets.** Fix $H : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $\det JH \equiv -3456$. The machine can output 0 in exactly three ways, $H^{-1}(0) = \{r_0, r_+, r_-\}$, but it can never output one chosen value Δ . Reading $\beta(x, y, z) = x^2$ turns those three inputs into one bit: r_0 means 0, while r_+ and r_- mean 1.
- 2. Use a Boolean value as a target switch.** Define $T_s(v) = H(v) - (1-s)\Delta$. Solving $T_s(v) = 0$ is the same as asking H to output $(1-s)\Delta$. If $s = 1$, this asks for 0, which is possible. If $s = 0$, it asks for Δ , which is impossible. Thus

$$\exists v : T_s(v) = 0 \iff s = 1 \quad (s \in \{0, 1\}).$$

- 3. Put one lock on every clause.** The equations $H(w_i) = 0$ force each variable to encode a bit. Let $s_j \in \{0, 1\}$ be the polynomial value of clause C_j , and add the equation $T_{s_j}(v_j) = 0$. The target-switch property says this equation has a solution exactly when C_j is true. Therefore all blocks vanish exactly when one assignment satisfies every clause.
- 4. The Jacobian remains constant.** Changing s_j only translates the output of H ; it does not change the derivative with respect to the fresh block v_j . The full Jacobian is block lower triangular with $n+m$ copies of JH on its diagonal, so $\det JK_\phi = (-3456)^{n+m}$ and $\deg K_\phi \leq 7$.